125101 - Analytical chemistry 1 for engineers

Tutorial 6 – Acids /bases and Buffer solutions

T6-1. Calculate the pH of the following solutions:

a. CH_3COOH solution (K_a=1.8×10^{-5}) with a concentration of 0.010 mol L^{-1};

b. $Cl_2CHCOOH$ solution (K_a=$10^{-1.26}$) with a concentration of 0.20 mol L^{-1}.

Note

pH is expressed to the 0.01 decimal place, regardless of what is justified by significant figures. We do this for the sake of consistency and also because 0.01 is near the limit of accuracy in pH measurements.

a.

$$fK_a = 0.010 \times 1.8 \times 10^{-5} = 1.8 \times 10^{-4} >> 10K_w\ (10^{-13})$$

$$\boldsymbol{f/K_a = 0.010/\ 1.8 \times 10^{-5} = 5.56 \times 10^{2} > 100}$$

So, the simple formula can be used,

$$\boxed{[H^+] = \sqrt{fKa}}$$

$= \sqrt{0.01 \times 1.8 \times 10^{-5}}$

$= 4.24 \times 10^{-4}$

pH = 3.37

If we use the following formula,

$$[H^+] = \frac{-Ka + \sqrt{Ka^2 + 4\,fKa}}{2}$$

$[H^+] = 4.15 \times 10^{-4}$, so, pH = 3.38

It is safe to use the simple formula to calculate the pH under $f/K_a > 100$.

b.

$$fK_a = 10^{-1.26} \times 0.20 = 0.011 >> 10K_w\,(10^{-13})$$

$$\boldsymbol{f/K_a = 0.20/10^{-1.26} = 3.64 < 100}$$

So, the simple formula **<u>cannot</u>** be used in this condition, the $[H^+]$ should be computed as following,

$$[H^+] = \frac{-Ka + \sqrt{Ka^2 + 4\,fKa}}{2}$$

$[H^+] = 0.080_9$, pH = -log$[H^+]$ =1.09

If you use the simple formula of $\boxed{[H^+] = \sqrt{fKa}}$

The results of $[H^+]$ is 0.10_5 and pH is 0.98, which is incorrect.

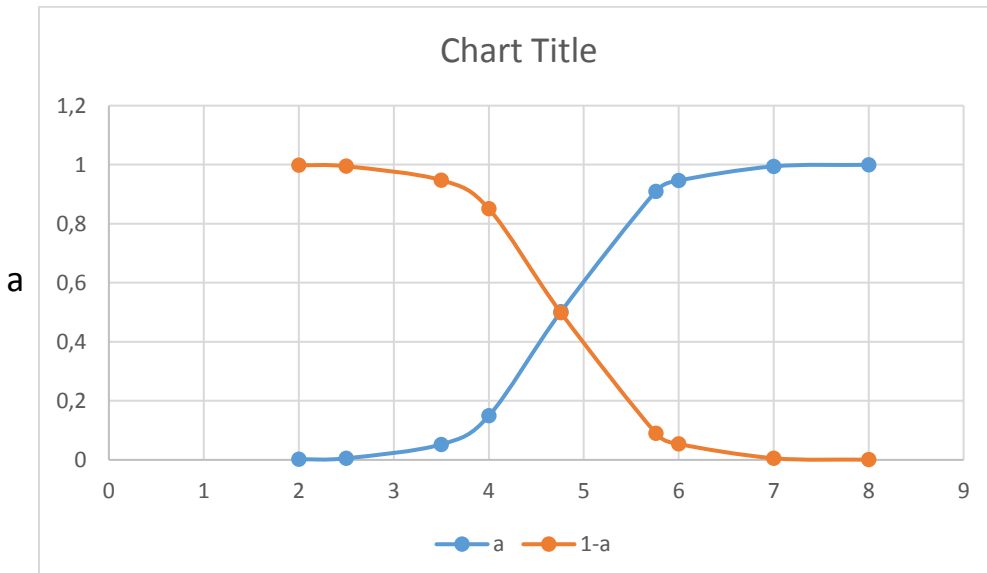
T6-2. Find the fraction of dissociation α of HAc (K_a=1.75×10^{-5}) at the following pHs:

pH	α
2.00	
2.50	
3.50	
4.00	
4.76	
5.76	
6.00	
7.00	
8.00	

$$a(Ac^-) = \frac{[Ac^-]}{[Ac^-]+[HAc]} \qquad k_a = \frac{[H^+][Ac^-]}{[HAc]}$$

$$\Longrightarrow a(Ac^-) = \frac{Ka}{[H^+]+Ka}$$

pH	a(Ac^-)	a(HAc)
2.00	0.00175	0.998
2.50	0.00550	0.995
3.50	0.0524	0.948
4.00	0.149	0.851
4.76	0.502	0.498
5.76	0.910	0.09
6.00	0.946	0.054
7.00	0.994	0.006
8.00	0.999	0.001



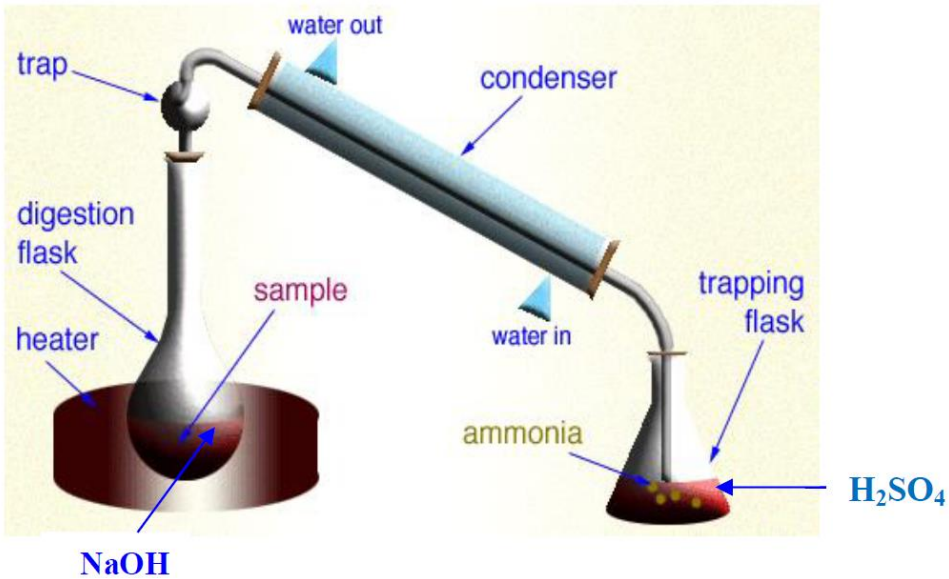
T6-3. The Kjeldahl method is an important method, in which we measure the amount of nitrogen in proteins, milk, cereals, etc.

In an experiment, 1.8 grams of ammonium decavanadate [$(NH_4)_6V_{10}O_{28}*6H_2O$] (1173.7 g/mol) were placed in a Kjeldahl container and cooked with an excess of NaOH to obtain ammonia. The ammonia was distilled and collected in a trapping flask containing 50 mL of 0.1M H_2SO_4. By the end of the reaction, the solution was titrated with 28 mL of 0.05 M NaOH.

What is the weight percentage of nitrogen in the sample? Compare the obtained value with the theoretical value from the formula weight.

$$(NH_4)_6V_{10}O_{28} * 6H_2O + 6NaOH \xrightarrow{\Delta} 6NH_3 + Na_6V_{10}O_{28} + 6H_2O$$

$$H_2SO_4\ (total) + 2NH_3 \rightarrow SO_4^{-2} + 2NH_4^+$$



Analyte $\rightleftharpoons$ NH_3 $\rightleftharpoons$ Excess strong acid

NaOH as titrant for the unreacted strong acid

Since there is an excess of the acid, part of the acid will react with the ammonia to form NH_4^+ and the rest will be titrated with NaOH via the following reaction.

$$H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + H_2O$$

The number of moles of the acid that reacted with ammonia is:

$$n(H_2SO_4) = 50\times10^{-3}\times0.1 - \frac{1}{2}\cdot28\times10^{-3}\times0.05 = 4.3\cdot10^{-3}$$

So, the number of moles, grams and the percentage of ammonia that were present are,

$$n(NH_3) = 2\times4.3\times10^{-3} = 8.6\times10^{-3}\,mol$$

$$m(N) = 8.6\times10^{-3}\times14 = 0.1204g$$

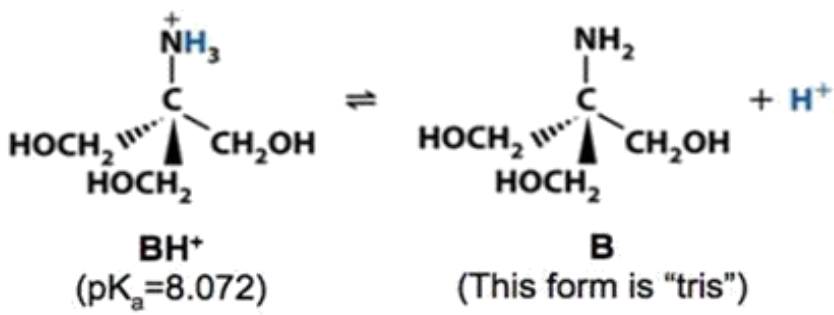
$$\%(N) = \frac{0.1204}{1.8}\times100\% = 6.69\%$$

From the calculation, the theoretical value is,

$$\%(N) = \frac{6 \times 14}{1173.7} \times 100\% = 7.16\%$$

It is demonstrated the measured result is close to the theoretical number.

T6-4. What is the pH of a buffer after adding 15.0 mL of 1.50 M HCl to 1.00L solution containing 12.43 gr of "tris" (hydroxymethyl-aminomethane, B) (121.135 gr/mole) and 4.67 gr of its conjugate acid BH^+ (157.596 gr/mol)? The tris pKa is 8.072.



The molarity for $[B]_o$ and $[BH^+]_o$ before HCl addition,

$$[B]_o = \frac{12.43\ g/L}{121.135\ g/mol} = 0.1026\ M$$

$$[BH^+]_o = \frac{4.67\ g/L}{157.596\ g/mol} = 0.0296\ M$$

The molarity for $[B]_f$ and $[BH^+]_f$ after HCl addition:

$$\mathbf{[B]_f = \frac{0.1026 mol/L \times 1L}{1\ L + 15.0\ mL} = 0.1011\ M}$$

$$\mathbf{[BH^+]_f = \frac{0.0296\ mol/L \times 1L}{1\ L + 15.0\ mL} = 0.02916\ M}$$

The molarity for HCl before adding,

$$\mathbf{[HCl] = \frac{1.50\ \frac{mol}{L} \times 15.0 mL}{1\ L + 15.0\ mL} = 0.02217\ M}$$

	B	**+ H^+**	**→ BH^+**
	Tris	from HCl	
Initial	0.1011	0.02217	0.02916
Final	0.07893	0	0.05133

The $[H^+]$ from HCl would be consumed by B and produce BH^+,

$$pH = pKa + \log \frac{[B]}{[BH^+]}$$

$$= 8.072 + log \frac{0.07893}{0.05133} = 8.26$$

Note, two approximations are employed to simplify the calculation,

Approximation-1

If you mix **A** moles of a weak acid with **B** moles of its conjugate base, the moles of acid remain close to A and the moles of base remain close to B. Little reaction occurs to change either concentration.

To understand why this should be so, consider an acid BH^+ with pKa = 8 and its conjugate base with pKb = 6. Let's calculate the fraction of acid dissociates in a 0.1 M solution.

$BH^+ \rightleftharpoons H^+ + B$

f-x *x* *x*

$\frac{x^2}{f-x} = Ka \Rightarrow x = 3.16\times10^{-3}$ M

So, fraction of dissociation $\alpha(B) = \frac{x}{f}$ 100% = 3.16%

The acid is only 3.16% dissociated under this conditions.

In a solution containing 0.03 M of B dissolved in 1.00 L, the extent of reaction of B with water is even small.

$B + H_2O \rightleftharpoons OH^- + BH^+$

f-y *y* *y*

$\frac{y^2}{f-y} = Kb \Rightarrow y = 1.7\times10^{-4}$ M

So, fraction of dissociation $\alpha(BH^+) = \frac{y}{f}$ 100% = 0.57%

BH^+ dissociates very little, and adding extra B to the solution makes BH^+ dissociate even less. Similar, B doesn't react much with water, and adding extra BH^+ makes B react even less. Therefore, 0.1026 mol of B plus 0.0296 mol of BH^+ are added to water, there will be close to 0.1026 mol of B and close to 0.0296 mol of BH^+ in solution at equilibrium.

Approximation-2

Strong acid reacts with a weak base essentially "completely" because the equilibrium constant is large.

$BH^+ \rightleftharpoons B + H^+$ Ka= 10^{-8}

If B is tris (hydroxymethyl) aminomethane, the equilibrium constant for reaction with HCl is

$B + HCl \rightleftharpoons BH^+Cl^-$

$K=\frac{1}{k_a} = 1.2\times10^8$

The equilibrium constant is very big. So, we can approximate the reaction percentage of the strong acid reacts with a weak base (or strong base reacts with a weak acid) reaches 100%.

In conclusion, the buffer pH doesn't change too much when a limited amount of strong acid is added. Addition of 12 mL of 1.0 M HCl changes from pH 8.61 to 8.41 ($\Delta pH = 0.2$). Addition of 12 mL of 1.0 M HCl to 1.0 L of unbuffered solution ($pH \approx 7$) would have lowered the pH to 1.93 ($\Delta pH \approx 5$).